SIMATS ENGINEERING

CO4 - ASSESSMENT TOOL 3

Article Review

COURSE NAME: Deep Learning **COURSE CODE: MLA0408**

COURSE OUTCOME COVERED

CO 4: Students will be able to analyze and evaluate advanced deep learning techniques and architectures, interpret research findings, and apply appropriate models to real-world problems. They will be able to critically review research articles, compare methodologies, and justify suitable deep learning solutions. (BL-6)

Assessment Tool Weightage: 30%

QUESTIONS:

Q1. Article Identification and Summary

Select a recent peer-reviewed research article related to Deep Learning, such as CNNs, Transfer Learning, ResNet, DenseNet, RNNs, LSTM, Autoencoders, or other advanced architectures. Provide the article title, authors, publication venue, year, and research objective. Summarize the main problem addressed, proposed approach, and key findings in your own words.

Q2. Problem Statement and Motivation

Explain the real-world problem addressed by the selected article and analyze why the problem requires a deep learning solution. Identify the limitations of existing approaches discussed by the authors and explain the motivation for the proposed method.

Q3. Methodology and Architecture Review

Critically review the methodology used in the article. Describe the dataset, preprocessing, model architecture, important layers or components, training strategy, and evaluation metrics. Illustrate the proposed architecture or workflow and explain the role of its major components.

Q4. Results, Comparison and Critical Analysis

Analyze the experimental results reported in the article. Compare the proposed method with the baseline or existing methods using the reported performance measures. Discuss the strengths, limitations, computational considerations, and validity of the results. Identify at least two possible improvements or future research directions.

Q5. Application and Personal Evaluation

Explain how the reviewed deep learning method can be applied to a practical industry problem. Propose a suitable implementation or extension and justify your choices. Conclude the review with your overall evaluation of the article, including its contribution, practical relevance, and scope for future work.

Article Review Submission Guidelines

- Choose one recent and relevant peer-reviewed research article in Deep Learning.
- Attach or provide the complete citation/reference of the selected article.
- The review must be written in the student's own words and should demonstrate critical analysis.
- Include architecture diagrams, tables, graphs, or screenshots where they improve explanation.
- Use appropriate academic referencing and avoid plagiarism.
- Recommended review length: 4–6 pages, excluding the reference page.

Code of Conduct:

I **G Vishnu Madhav(192425192)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developin g	Level 1 – Beginning
Article Understanding & Summary	20%	Comprehensive and accurate understanding of the article, problem, objectives, and contributions.	Good understanding with minor omissions or inaccuracies.	Basic understanding; important details are missing.	Poor understanding or significant misinterpretation .
Critical Review of Methodology	25%	Thoroughly evaluates methodolog, architecture, datasets, training, and evaluation choices.	Appropriate analysis with minor gaps.	Limited analysis with several unexplained aspects.	Little or no meaningful analysis of methodology.
Analysis of Results	20%	Accurately interprets results, comparison s, metrics, strengths, and limitations.	Good interpretation with reasonable comparison.	Basic interpretation with limited comparison.	Incorrect, unsupported, or missing interpretation.
Application & Proposed Improvements	20%	Practical, innovative application with well-justified improvements and future directions.	Relevant application and feasible improvements.	Basic application with limited justification.	Unclear, impractical, or unsupported proposal.
Presentation & Referencing	15%	Well-organized, professional presentation with clear diagrams and	Clear presentatio n with minor formatting	Adequate presentation with	Poor organization, unclear presentation,

		correct academic referencing.	or referencing issues.	inconsistent formatting/refe ren cing.	or inadequate referencing.

Suggested Article Review Structure

1. Article Citation
2. Abstract / Executive Summary
3. Research Problem and Motivation
4. Methodology and Deep Learning Architecture
5. Dataset and Experimental Setup
6. Results and Comparative Analysis
7. Strengths and Limitations
8. Proposed Improvements / Future Scope
9. Practical Industry Application
10. Conclusion and References

1. Article Citation

Dharmik, A. (2025). COVID-19 Pneumonia Diagnosis Using Medical Images: Deep Learning-Based Transfer Learning Approach. JMIRx Med, 6, e75015. https://doi.org/10.2196/75015 (also available as arXiv:2503.12642).

Author: Anjali Dharmik, Senior AI Engineer, HP Inc.

Venue / Year: JMIRx Med (peer-reviewed), 2025

Research Objective: To build and evaluate a transfer-learning-based CNN system that can quickly and accurately diagnose COVID-19 pneumonia from chest X-ray and CT images, in a way that stays reliable as new virus variants appear.

2. Abstract / Executive Summary

The article addresses the problem of diagnosing COVID-19 pneumonia quickly and reliably from medical images, at a point where the disease is no longer a global emergency but still burdens healthcare systems, particularly for high-risk patients. The author frames diagnostic imaging as a faster, more scalable alternative to traditional lab-based testing.

The proposed approach fine-tunes three well-known pre-trained CNN architectures — VGG16, VGG19, and ResNet50 — on a combined dataset of 6,259 chest X-ray and CT images (4,651 COVID-19 cases and 1,608 normal cases). Images are converted to grayscale, resized to 150×150 pixels, augmented, and normalized before being fed into each network for binary classification (COVID-19 vs normal).

The key finding is that all three architectures, once properly tuned, converge to nearly identical peak performance: 97.73% accuracy, 100% sensitivity, 93.33% specificity, and a 98% F1-score. ResNet50 reaches this peak through clear, incremental improvement as hyperparameters are tuned, while VGG16 plateaus early (a sign of underfitting) and VGG19 improves steadily across iterations.

3. Research Problem and Motivation

The real-world problem is straightforward but high stakes: clinicians need a way to tell COVID-19 pneumonia apart from a normal chest scan quickly, cheaply, and at scale, especially in settings where laboratory testing (such as RT-PCR or serology) is slow, expensive, or in short supply. The author notes that traditional diagnostic methods suffer from low sensitivity and long turnaround times, while chest X-rays and CT scans are already part of routine respiratory workups and can reveal COVID-19-related lung changes, sometimes before symptoms appear.

This is a genuine deep learning problem rather than a simple rules-based one because the visual difference between COVID-19 pneumonia and other causes of lung opacity is subtle, varies from patient to patient, and is hard to capture with hand-crafted image features. Deep CNNs can learn these patterns directly from pixel data given enough labeled examples, which is why the field has moved toward automated image-based screening.

The author's motivation for using transfer learning specifically is data scarcity: training a large CNN from scratch needs far more labeled medical images than are typically available. By starting from networks already pre-trained on large natural-image datasets (ImageNet) and fine-tuning them on the smaller COVID-19 dataset, the model can reuse general visual features (edges, textures, shapes) and only needs to relearn the task-specific patterns, reducing both the data and compute required. The article also points to a limitation of prior studies — inconsistent or under-explored hyperparameter tuning — as a gap the current work tries to close by systematically varying learning rate, batch size, dropout, and epoch count.

4. Methodology and Deep Learning Architecture

4.1 Dataset and Preprocessing

The study combines two public datasets into 6,259 chest images: 4,651 COVID-19-positive images and 1,608 normal images, drawn from Kaggle chest X-ray/CT collections. Preprocessing steps include grayscale conversion, resizing every image to 150×150 pixels, and noise reduction to remove artifacts common in scanned medical images. Data augmentation (rotation, flipping, colour-space changes, and kernel filtering) is applied to reduce overfitting given the limited dataset size, and images are split 80% training, 10% validation, and 10% testing.

4.2 Model Architecture

Three pre-trained CNN backbones are fine-tuned for binary classification (COVID-19 = 1, Normal = 0):

- VGG16 / VGG19 — 16 and 19-layer networks built from stacked small 3×3 convolution filters; simple, uniform architecture that is easy to fine-tune but prone to underfitting on small datasets.
- ResNet50 — a 50-layer network that uses residual (skip) connections to counter the vanishing-gradient problem, allowing much deeper networks to train effectively.

Each backbone's convolutional base supplies pre-learned visual features, while the final classification layers are retrained on the COVID-19 dataset. Regularization is handled through dropout (0.2–0.5) and batch normalization to control overfitting.

4.3 Training Strategy and Evaluation Metrics

Hyperparameters were swept systematically: learning rate between 0.0001 and 0.01, batch size of 8 or 16, dropout between 0.2 and 0.5, and epoch counts of 50 or 100. Each configuration was evaluated with repeated cross-validation (producing 15 learning-curve and confusion-matrix pairs across the three models), and performance was scored using accuracy, sensitivity (recall), specificity, and F1-score rather than accuracy alone — an appropriate choice for a medical screening task where missing a positive case (false negative) is far costlier than a false alarm.

5. Results and Comparative Analysis (Q4)

Model	Accuracy	Sensitivity	Specificity	F1-Score
VGG16 (best setting)	97.73%	100%	93.33%	0.98
VGG19 (best setting)	97.73%	100%	93.33%	0.98
ResNet50 (best setting)	97.73%	100%	93.33%	0.98
ResNet50 (first tuning pass)	90.91%	86.21%	100.00%	0.91

All three architectures reach the same ceiling performance (97.73% accuracy, 100% sensitivity, 93.33% specificity, 98% F1-score) once hyperparameters are well tuned, but they get there differently. ResNet50's accuracy rises steadily across tuning iterations (90.91% → 93.18% → 95.45% → 97.73%), which is consistent with its residual connections allowing it to keep learning as training becomes more thorough. VGG19 shows a similar upward trend. VGG16, by contrast, reports the same metrics across every hyperparameter combination tested, which the author interprets as a sign of underfitting rather than genuine convergence — the model may be too shallow or too quickly saturated to benefit further from tuning on this dataset size.

6. Strengths

- 100% sensitivity in the best configurations means, on this test set, no COVID-19 case was missed — the most important property for a screening tool.
- Comparing three architectures under identical preprocessing and cross-validation gives a fair, controlled comparison rather than a single cherry-picked model.
- Reporting sensitivity, specificity, and F1-score alongside accuracy avoids the classic pitfall of overstating performance on an imbalanced dataset (COVID-19 cases outnumber normal cases roughly 3:1).

7. Limitations

- The dataset is class-imbalanced (4,651 vs 1,608) and drawn from a small number of public sources, so results may not generalize to other hospitals, scanners, or populations.
- Binary classification (COVID-19 vs normal) does not distinguish COVID-19 pneumonia from other types of pneumonia, which is the more clinically realistic and harder problem.
- The identical metrics across all VGG16 runs suggest either underfitting or a coarse-grained evaluation that cannot detect real differences between configurations — this weakens confidence in the reported numbers for that model.
- No external, held-out dataset was used to confirm the results outside the original data source, and computational cost/inference-time figures are not reported, which matters for real deployment.

8. Proposed Improvements / Future Research Directions

- Multi-class classification across several respiratory diseases (COVID-19, tuberculosis, bacterial pneumonia, normal), which the author already flags as future work, would better match real clinical decision-making.
- External validation on an independent multi-hospital dataset, combined with class-imbalance-aware training (e.g., focal loss or class weighting) rather than augmentation alone, would give a more trustworthy estimate of real-world performance.

9. Practical Industry Application and Personal Evaluation

9.1 Practical Industry Application

This approach fits naturally into a clinical decision-support tool: a triage-assist system where every chest X-ray or CT taken in an emergency department or radiology unit is automatically pre-screened by the fine-tuned ResNet50 model, flagging likely COVID-19 cases for a radiologist's priority review rather than replacing the radiologist. Given the very high sensitivity, it could also be extended to a low-resource or rural telemedicine setting, where a technician captures an X-ray and a cloud- or edge-deployed model gives an immediate second opinion while a specialist review is arranged.

As an extension, I would propose adding an explainability layer (e.g., Grad-CAM heatmaps) so clinicians can see which lung regions drove the prediction, and packaging the model as a lightweight, quantized version (using MobileNet-style distillation) so it can run on-device in resource-constrained clinics rather than requiring cloud inference for every image.

9.2 Overall Evaluation

Overall, this is a solid, well-structured applied deep learning study. Its main contribution is a careful, side-by-side comparison of three transfer-learning backbones under a consistent preprocessing and evaluation pipeline, showing that transfer learning can reach near-perfect sensitivity on this dataset with modest data and compute. Its practical relevance is high for imaging-based triage, though the binary framing, class imbalance, and lack of external validation limit how far the reported numbers can be trusted outside this specific dataset. The clearest scope for future work — multi-class diagnosis and independent validation — is one the author already identifies, and addressing it would turn this from a promising proof of concept into a deployable clinical tool.

10. Conclusion and References

This review examined Dharmik (2025), a peer-reviewed transfer-learning study that fine-tunes VGG16, VGG19, and ResNet50 for COVID-19 detection from chest X-ray and CT images. The study demonstrates that transfer learning can achieve strong diagnostic performance (up to 97.73% accuracy and 100% sensitivity) even on a moderately sized medical imaging dataset, while also revealing architecture-specific behaviour (ResNet50's steady improvement versus VGG16's underfitting) that is useful for practitioners choosing a backbone for similar tasks.

Reference: Dharmik, A. (2025). COVID-19 Pneumonia Diagnosis Using Medical Images: Deep Learning-Based Transfer Learning Approach. JMIRx Med, 6, e75015. https://doi.org/10.2196/75015.

Certification: I certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

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Signature: